Evaluation of Reconfigurable multiplier based Hardware accelerators for DNNs

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MOTIVATION & OVERVIEW

- Deep neural networks (DNNs) are widely used in vision, NLP, and autonomous systems. However, deploying these models on edge devices is constrained by strict energy and compute limits.
- Most accelerators use fixed-width multipliers, wasting power when layers need lower precision.
- Precision-gated radix-4 serial multiplier: Runtime-configurable 3–16 bit FP multiplier using a precision-gated Radix-4 Booth core that executes only the cycles required by the assigned mantissa width.

PROCESS FLOW

NSGA-II Optimizer — *Precision assignment per layer*

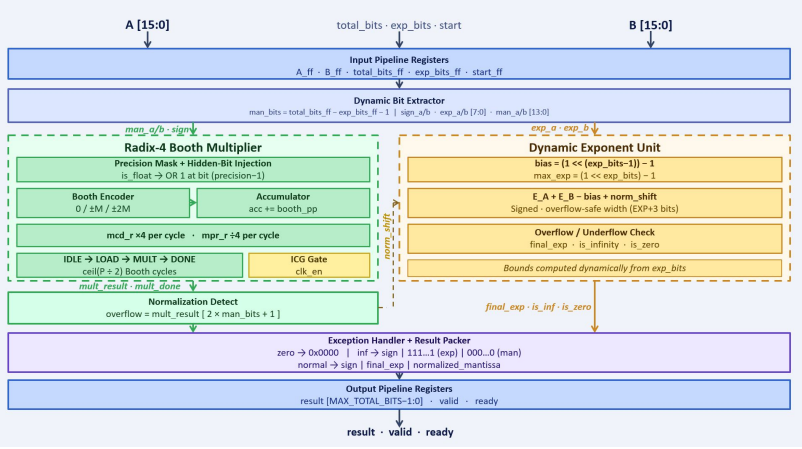
Reconfig FP Multiplier — *Radix-4 Booth Dynamic exp unit*

Genus RTL Synthesis — *Power · Area Latency*

Timeloop Simulator — *Energy · TOPS/W per layer*

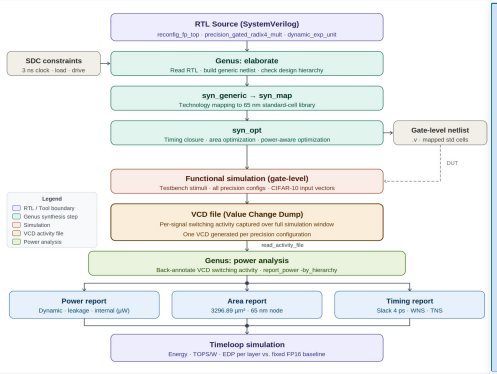
End-to-end co-design: optimization → hardware synthesis → architecture simulation

PRECISION-GATED RADIX-4 SERIAL RECONFIGURABLE MULTIPLIER DESIGN



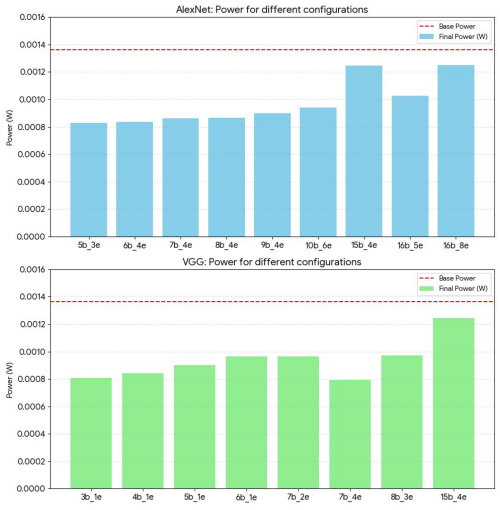
"A single synthesized multiplier that reconfigures its floating-point precision per layer at runtime; no additional hardware, no re-synthesis, no design overhead"

CADENCE GENUS AND TIMELOOP TOOL FLOW



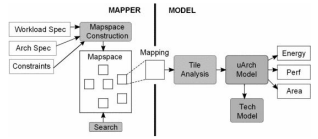
RTL designs were synthesized in Cadence Genus (65 nm) to obtain a fixed hardware implementation. Switching activity from gate-level simulations was captured using VCD files to compute configuration dependent energy values which were integrated into the Gemmini accelerator model in Timeloop to evaluate system-level efficiency under DNN workloads.

GENUS POWER RESULTS FOR DIFFERENT CONFIGURATIONS

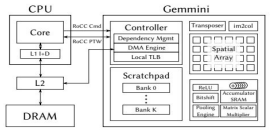


Reduced-precision configurations show consistently lower power than FP16, confirming that datapath switching activity is strongly dependent on mantissa width, which dominates multiplier complexity. Power does not scale strictly linearly with total bit-width, indicating that internal normalization and data dependent switching behavior also influence energy. Overall, reducing mantissa precision provides the most significant power savings, while exponent variation has comparatively smaller impact.

TIMELOOP AND GEMMINI HARDWARE ACCELERATOR

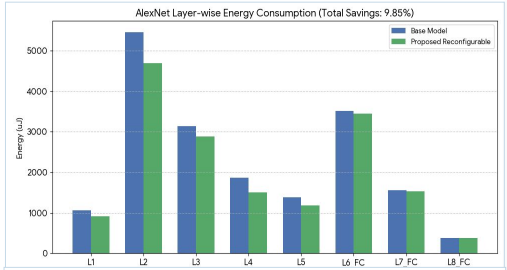


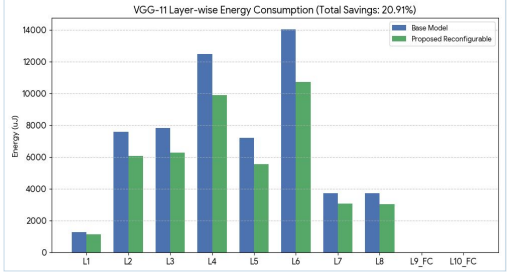
Timeloop tool-flow



Gemmini hardware architectural template

TIMELOOP LAYERWISE ENERGY CONSUMPTION RESULTS





Energy savings are consistently observed across workloads, with 20.91% reduction for VGG-11 and 9.85% for AlexNet, highlighting the effectiveness of precision reconfigurability at the system level. The relatively smaller gain in AlexNet is attributable to its lower compute intensity and higher proportion of memory access energy, which limits the impact of multiplier optimization. Overall, the results reinforce that benefits scale with MAC density, with compute dominant convolution layers deriving the greatest advantage from reduced-precision arithmetic.

KEY PERFORMANCE METRICS

~10-21%	3 ns	< 1%	8000+
Energy Reduction	Target Clock Period	Accuracy Drop	Floating point configurations supported

CONCLUSION AND FUTURE PROSPECTS

The proposed reconfigurable multiplier, evaluated on the Gemmini hardware accelerator, achieves significant energy reductions of 20.91% for VGG-11 and 9.85% for AlexNet. The hardware-software co-design maintains high performance with an accuracy drop of less than 1% and a 3 ns target clock period, while occupying a 33.4% area overhead at the 65nm technology node.

Future work will focus on scaling the optimization framework to more complex transformer-based architectures and larger datasets to further validate runtime precision scaling benefits. We plan to evaluate the model across a wider range of hardware accelerators and investigate implementation on advanced technology nodes (sub-65nm) to further minimize the current area footprint.